Chapter 12 Notes

Section 0

Notes

Note that when the relation $x = y$ is used (for instnace in P_1), this is a substitute for the formula

$$\forall z(z \in x \equiv z \in y)$$

Section 1

Exercise 1.1

We show that for finite α, $L_\alpha = R_\alpha$.

Proof. By induction. $L_0 = R_0$ by definition. Suppose $L_n = R_n$ where n is finite. By C_0, L_n is transitive (and thus also extensional), and hence by P_3, $\mathcal{F}(L_n) = \mathcal{P}(L_n) = \mathcal{P}(R_n) = R_{n+1}$, therefore completing the induction. □

Exercise 1.2

By Exercise 1.1, for finite α, $L_\alpha = R_\alpha$. Then it obviously follows that $L_\omega = R_\omega$. However, $L_{\omega+1} \neq R_{\omega+1}$. For recall that, if a set A is infinite, then $\mathcal{F}(A) \preceq A$. Thus $L_{\omega+1} = \mathcal{F}(L_\omega) \preceq L_\omega$ (because L_ω is infinite). But $R_{\omega+1} = \mathcal{P}(R_\omega) \succ R_\omega$ ($\succ L_\omega$) by Cantor's Theorem. Hence $L_{\omega+1} \neq R_{\omega+1}$, and $R_{\omega+1} \succ L_{\omega+1}$.

Section 2

Notes

It seems that ϕ being absolute over K is in fact equivalent to saying that K reflects V with respect to ϕ.

Note the following fallacious argument: "In fact, after showing that $x \in y$ is absolute, it immediately follows that $x \in y^+$ is absolute, as it is a special case of $x \in y$ being absolute." This fails because y^+ is not a variable.

It is remarked in (15) that "We already know that the condition 'x is transitive' is absolute over any transitive class". I do not know where we established this before this exercise.

The formula provided in (26) does not seem correct. For, in fact, under this definition, it would follow that all ordered pairs were unordered pairs. Because, for any ordered pair $x : \langle y, z\rangle = \{\{y\}, \{y, z\}\}$, there are some w and v such

that $x = \{w, v\}$, namely when $w = \{y\}$ and $v = \{y, z\}$ (or the other way around). A better definition would be, intuitively, x is an ordered pair iff there are no elements y, z such that $x = \langle y, z\rangle$, or, formally:

$$(\forall y \in x)(\forall z_1 \in y)(\forall z_2 \in y)(x \neq \langle z_1, z_2\rangle)$$

The formula given for (29) does not seem correct to me. It needs to show that there are no $\langle y, z_1\rangle \in x$ and $\langle y, z_2\rangle \in x$ where $z_1 \neq z_2$, yet it does not guarantee this. Instead, we can define (29) using (30) and (31). x is a function iff

$$x \text{ is a relation} \wedge (\forall y \in \mathrm{Dom}(x))(\forall z_1 \in \mathrm{Ran}(x))(\forall z_2 \in \mathrm{Ran}(x))$$

$$((\langle y, z_1\rangle \in x \wedge \langle y, z_2\rangle \in x) \supset z_1 = z_2)$$

Here is the proof of (31). First, note that $y \in \mathrm{Ran}(x)$ is Σ_0, as $y \in \mathrm{Ran}(x)$ iff $(\exists z \in x)(\exists w \in z)(\exists v \in w)(z = \langle v, y\rangle)$. Then $y = \mathrm{Ran}(x)$ iff

$$(\forall z \in y)(z \in \mathrm{Ran}(x)) \wedge (\forall z_1 \in x)(\forall z_2 \in z_1)(\forall z_3 \in z_2)$$

$$(z_3 \in \mathrm{Dom}(x) \subset z_3 \in y)$$